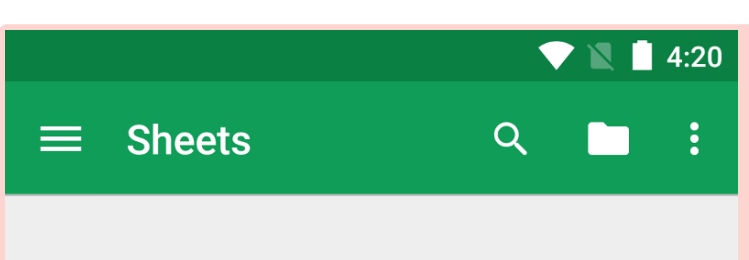
Menu & Dialog

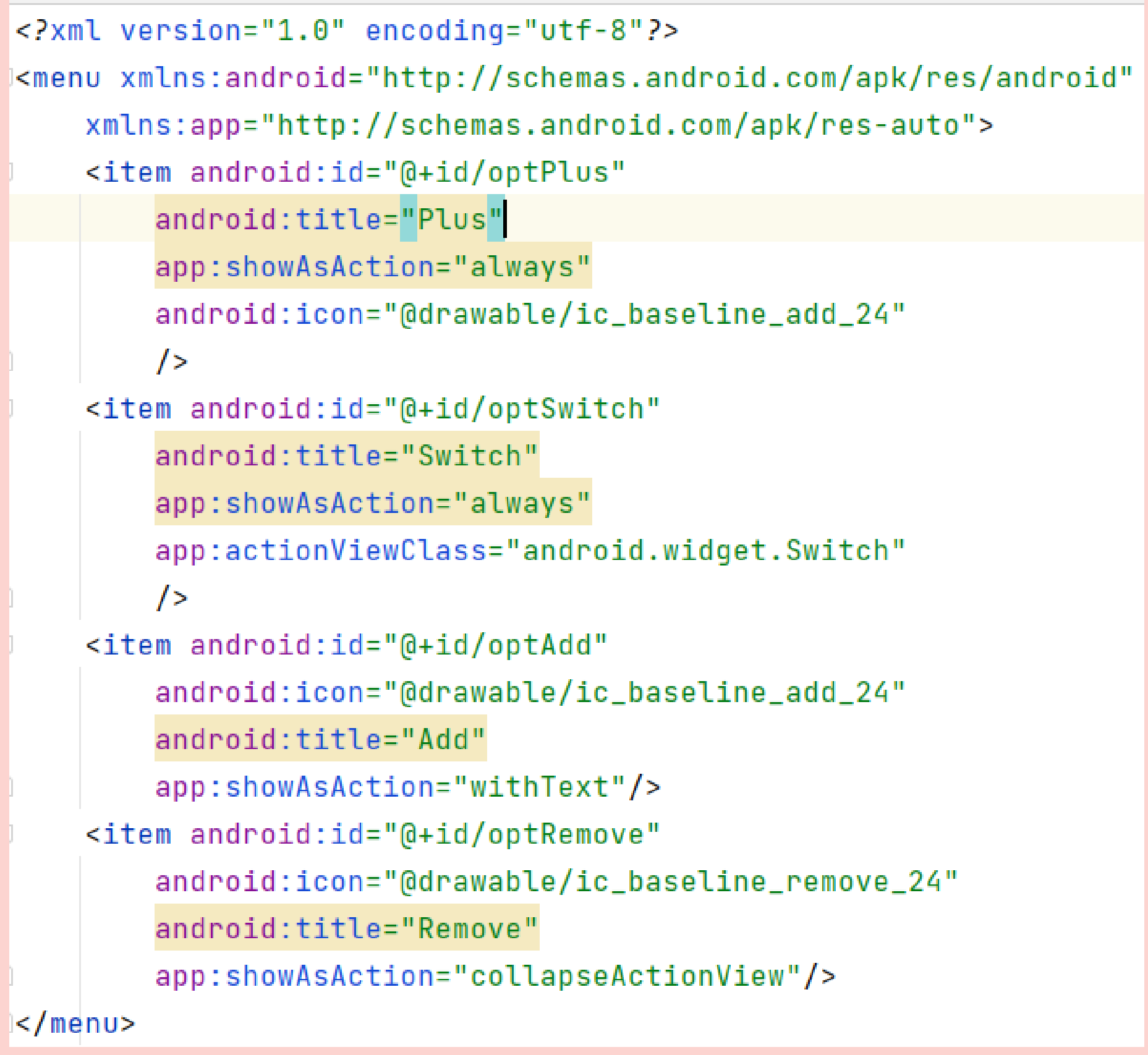
Menu

Option Menu



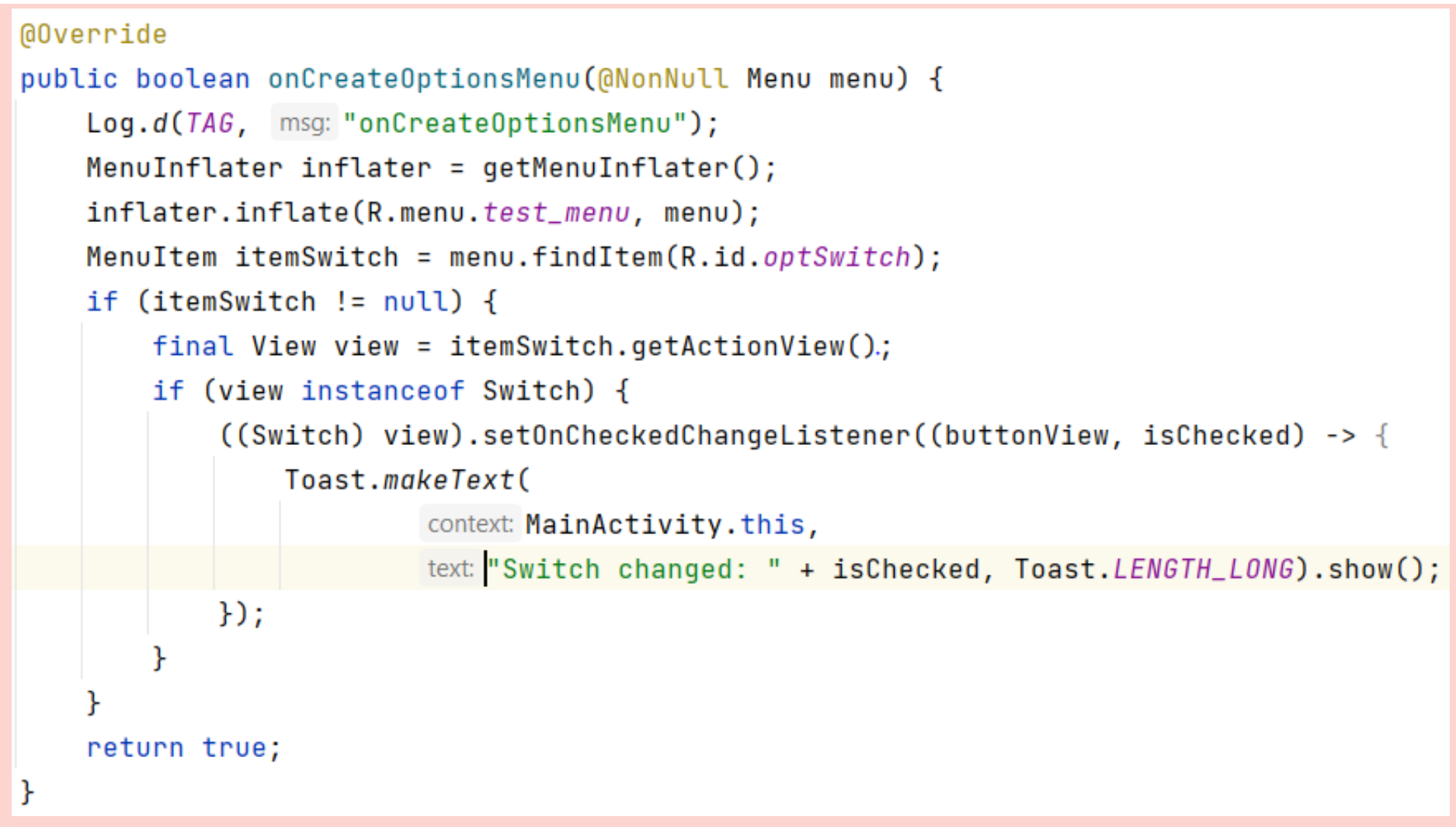
Look

Implementation



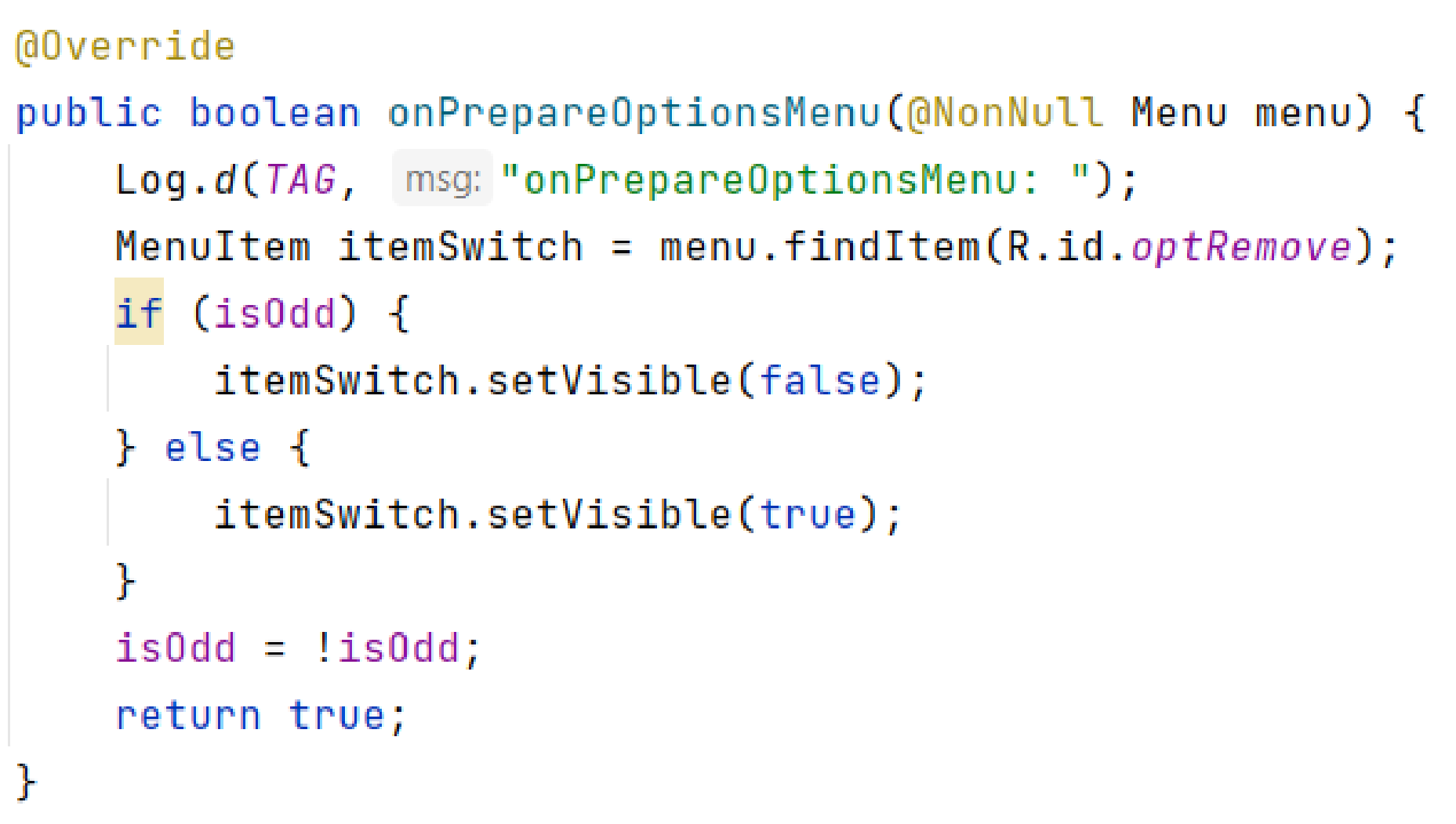
```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item android:id="@+id/optPlus"
        android:title="Plus"
        app:showAsAction="always"
        android:icon="@drawable/ic_baseline_add_24"
        />
    <item android:id="@+id/optSwitch"
        android:title="Switch"
        app:showAsAction="always"
        app:actionViewClass="android.widget.Switch"
        />
    <item android:id="@+id/optAdd"
        android:icon="@drawable/ic_baseline_add_24"
        android:title="Add"
        app:showAsAction="withText"/>
    <item android:id="@+id/optRemove"
        android:icon="@drawable/ic_baseline_remove_24"
        android:title="Remove"
        app:showAsAction="collapseActionView"/>
</menu>
```

① Define menu XML



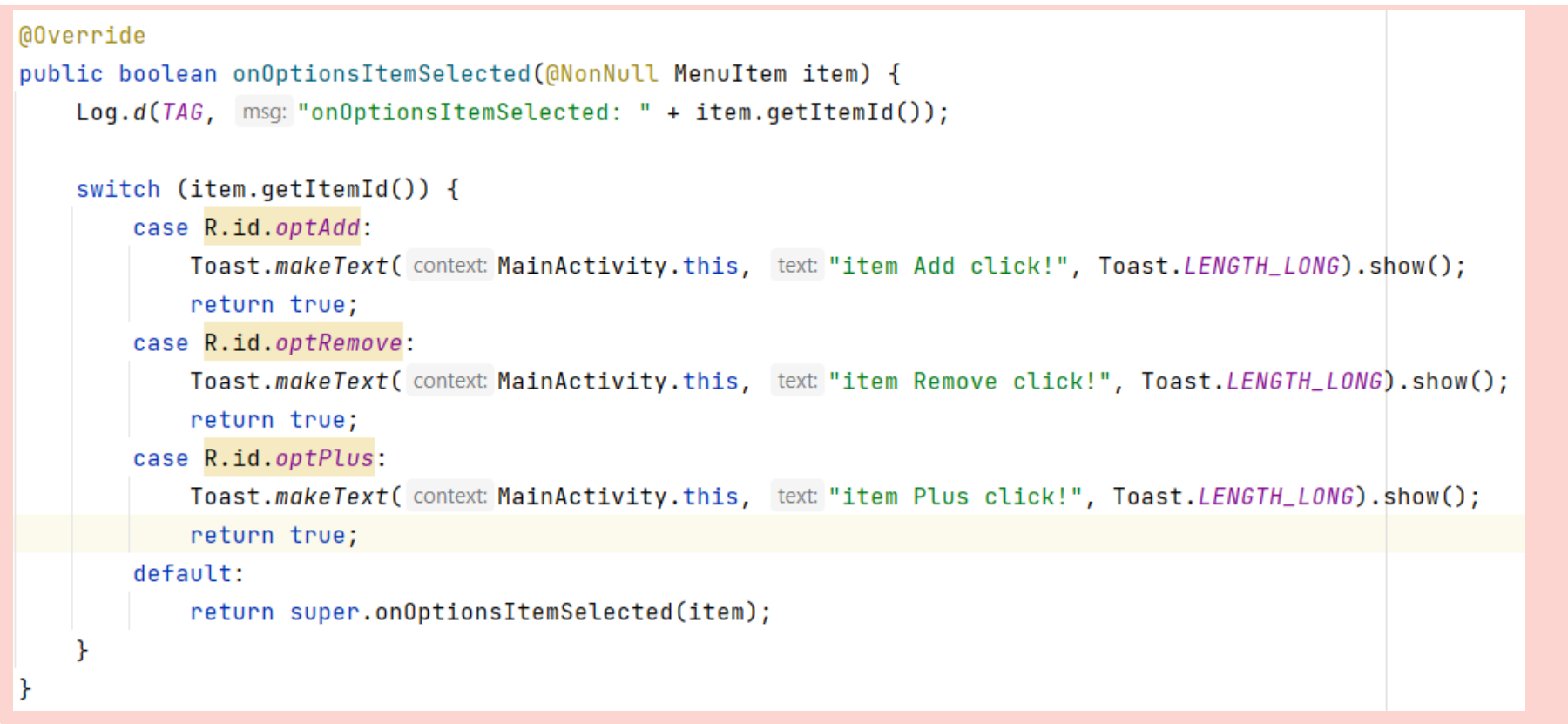
```java
@Override
public boolean onCreateOptionsMenu(@NonNull Menu menu) {
    Log.d(TAG, msg: "onCreateOptionsMenu");
    MenuInflater inflater = getMenuInflater();
    inflater.inflate(R.menu.test_menu, menu);
    MenuItem itemSwitch = menu.findItem(R.id.optSwitch);
    if (itemSwitch != null) {
        final View view = itemSwitch.getActionView();
        if (view instanceof Switch) {
            ((Switch) view).setOnCheckedChangeListener((buttonView, isChecked) -> {
                Toast.makeText(
                        context: MainActivity.this,
                        text: "Switch changed: " + isChecked, Toast.LENGTH_LONG).show();
            });
        }
    }
    return true;
}
```

② Implement **onCreateOptionMenu()** method



```java
@Override
public boolean onPrepareOptionsMenu(@NonNull Menu menu) {
    Log.d(TAG, msg: "onPrepareOptionsMenu: ");
    MenuItem itemSwitch = menu.findItem(R.id.optRemove);
    if (isOdd) {
        itemSwitch.setVisible(false);
    } else {
        itemSwitch.setVisible(true);
    }
    isOdd = !isOdd;
    return true;
}
```

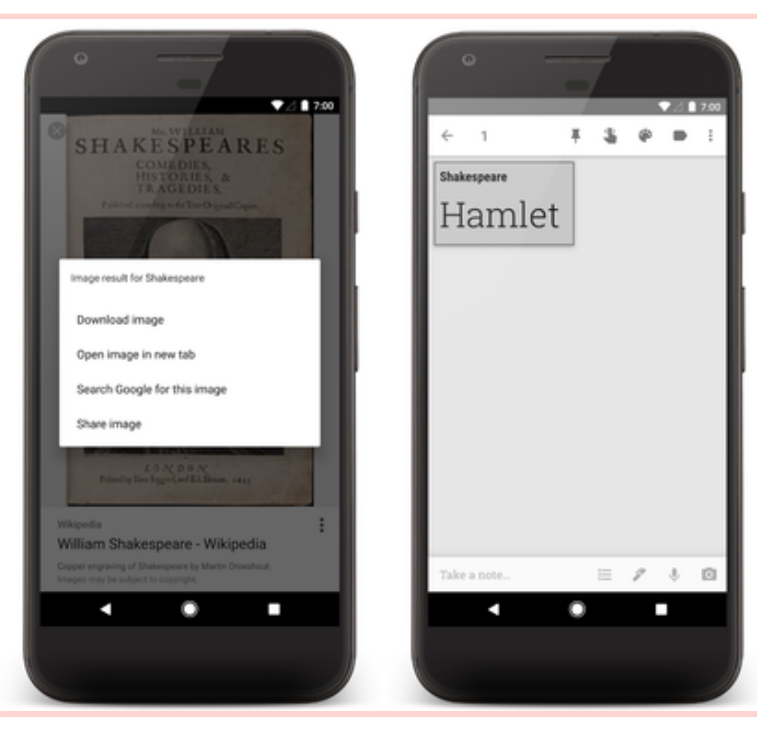
③ Implement **onPrepareOptionsMenu()** method (**Optional**)



```java
@Override
public boolean onOptionsItemSelected(@NonNull MenuItem item) {
    Log.d(TAG, msg: "onOptionsItemSelected: " + item.getItemId());

    switch (item.getItemId()) {
        case R.id.optAdd:
            Toast.makeText( context: MainActivity.this,  text: "item Add click!", Toast.LENGTH_LONG).show();
            return true;
        case R.id.optRemove:
            Toast.makeText( context: MainActivity.this,  text: "item Remove click!", Toast.LENGTH_LONG).show();
            return true;
        case R.id.optPlus:
            Toast.makeText( context: MainActivity.this,  text: "item Plus click!", Toast.LENGTH_LONG).show();
            return true;
        default:
            return super.onOptionsItemSelected(item);
    }
}
```

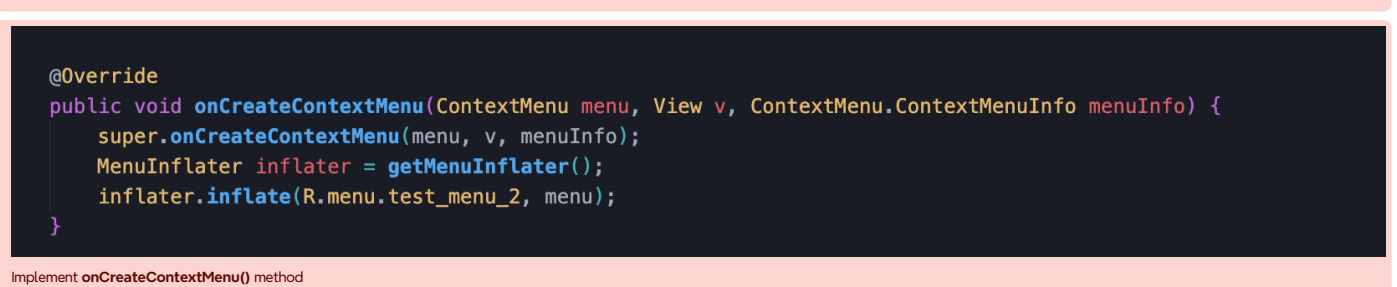
④ **Handle click event item menu** listener

Contextual Menu


SHAKESPEARES
Image result for Shakespeare
Download image
Open image in new tab
Search Google for this image
Share image
William Shakespeare - Wikipedia
Shakespeare
Hamlet

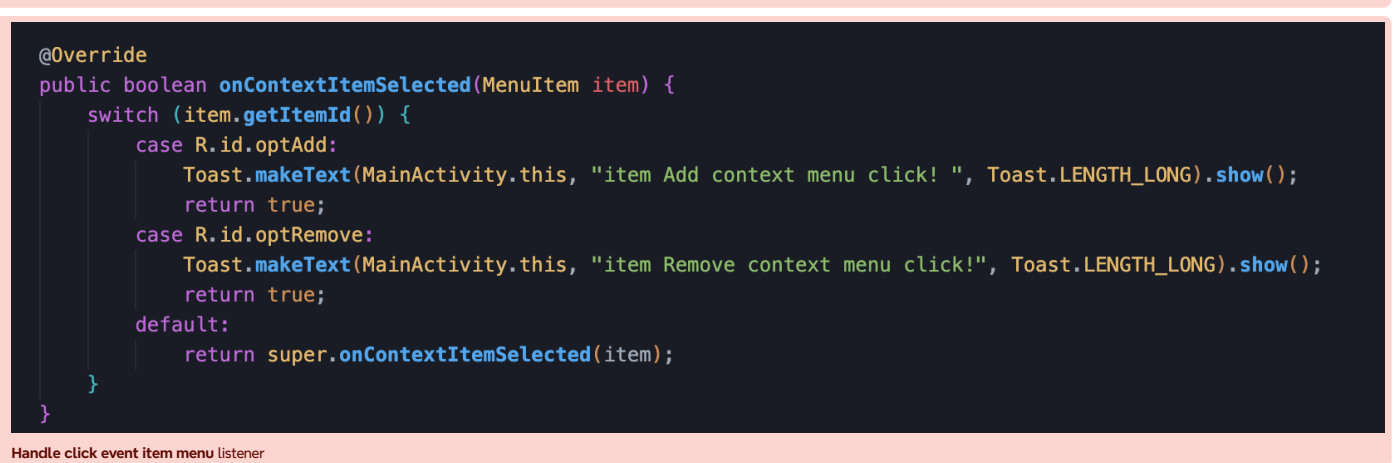
Floating contextual menu (img 1)

① **Define Menu XML**



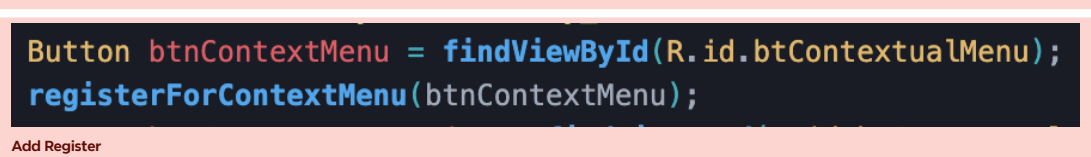
```java
@Override
public void onCreateContextMenu(ContextMenu menu, View v, ContextMenu.ContextMenuInfo menuInfo) {
    super.onCreateContextMenu(menu, v, menuInfo);
    MenuInflater inflater = getMenuInflater();
    inflater.inflate(R.menu.test_menu_2, menu);
}
```

② Implement **onCreateContextMenu()** method



```java
@Override
public boolean onContextItemSelected(MenuItem item) {
    switch (item.getItemId()) {
        case R.id.optAdd:
            Toast.makeText(MainActivity.this, "item Add context menu click! ", Toast.LENGTH_LONG).show();
            return true;
        case R.id.optRemove:
            Toast.makeText(MainActivity.this, "item Remove context menu click!", Toast.LENGTH_LONG).show();
            return true;
        default:
            return super.onContextItemSelected(item);
    }
}
```

③ **Handle click event item menu** listener

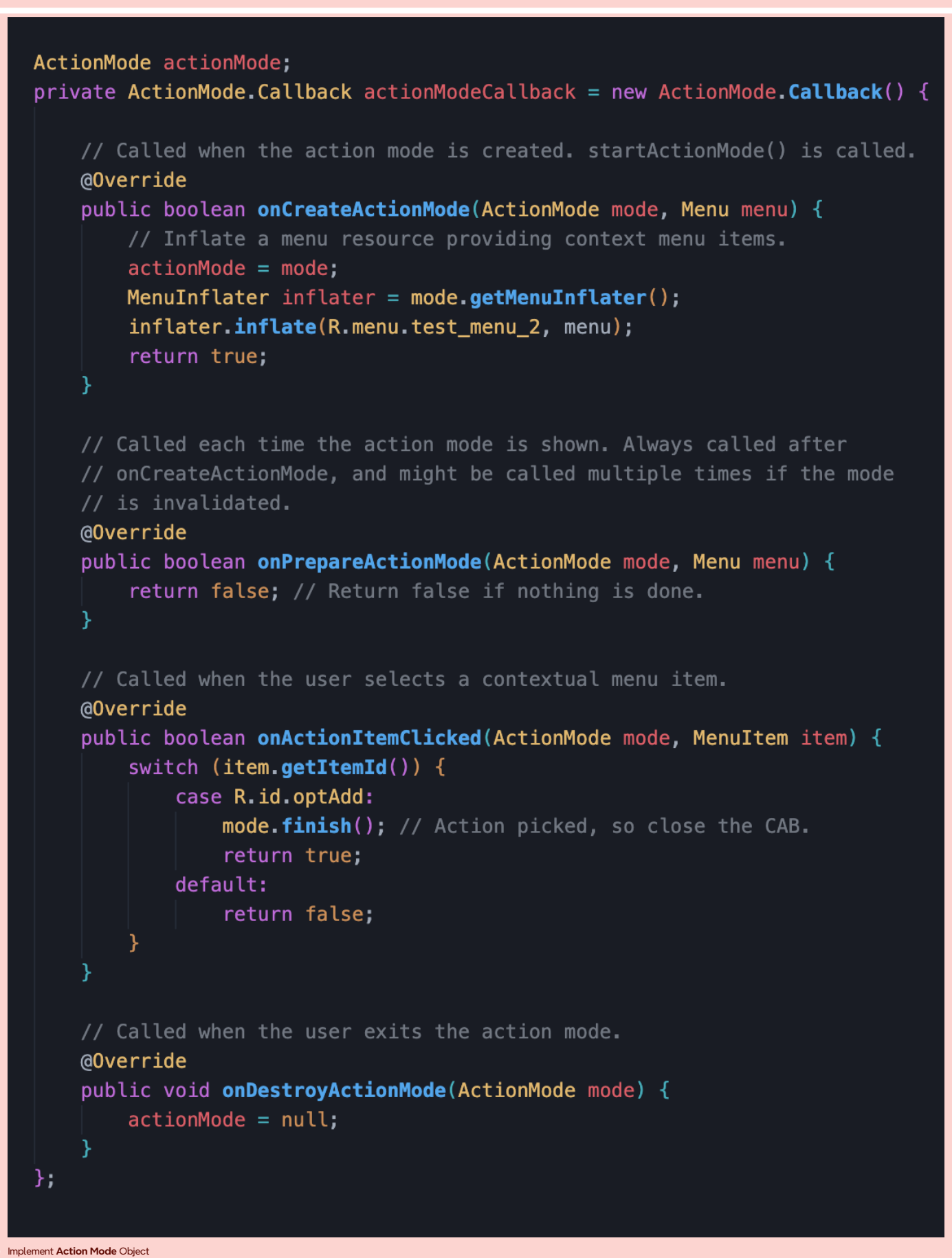


```java
Button btnContextMenu = findViewById(R.id.btContextualMenu);
registerForContextMenu(btnContextMenu);
```

④ **Add Register**

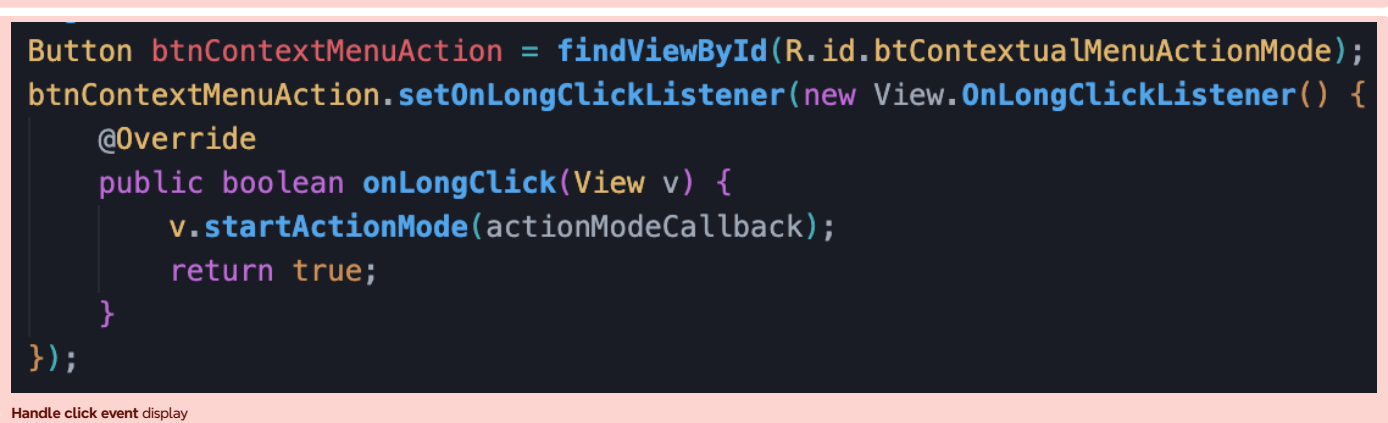
Action Context menu (img 2)

① **Define menu XML**



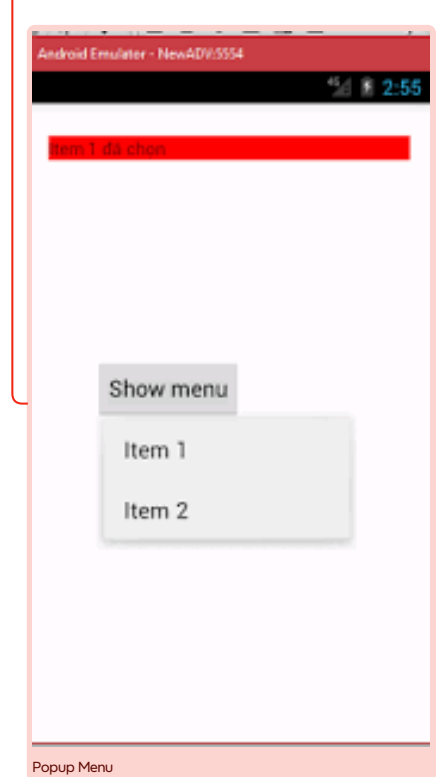
```java
ActionMode actionMode;
private ActionMode.Callback actionModeCallback = new ActionMode.Callback() {

    // Called when the action mode is created. startActionMode() is called.
    @Override
    public boolean onCreateActionMode(ActionMode mode, Menu menu) {
        // Inflate a menu resource providing context menu items.
        actionMode = mode;
        MenuInflater inflater = mode.getMenuInflater();
        inflater.inflate(R.menu.test_menu_2, menu);
        return true;
    }

    // Called each time the action mode is shown. Always called after
    // onCreateActionMode, and might be called multiple times if the mode
    // is invalidated.
    @Override
    public boolean onPrepareActionMode(ActionMode mode, Menu menu) {
        return false; // Return false if nothing is done.
    }

    // Called when the user selects a contextual menu item.
    @Override
    public boolean onActionItemClicked(ActionMode mode, MenuItem item) {
        switch (item.getItemId()) {
            case R.id.optAdd:
                mode.finish(); // Action picked, so close the CAB.
                return true;
            default:
                return false;
        }
    }

    // Called when the user exits the action mode.
    @Override
    public void onDestroyActionMode(ActionMode mode) {
        actionMode = null;
    }
};
```

② Implement **Action Mode** Object



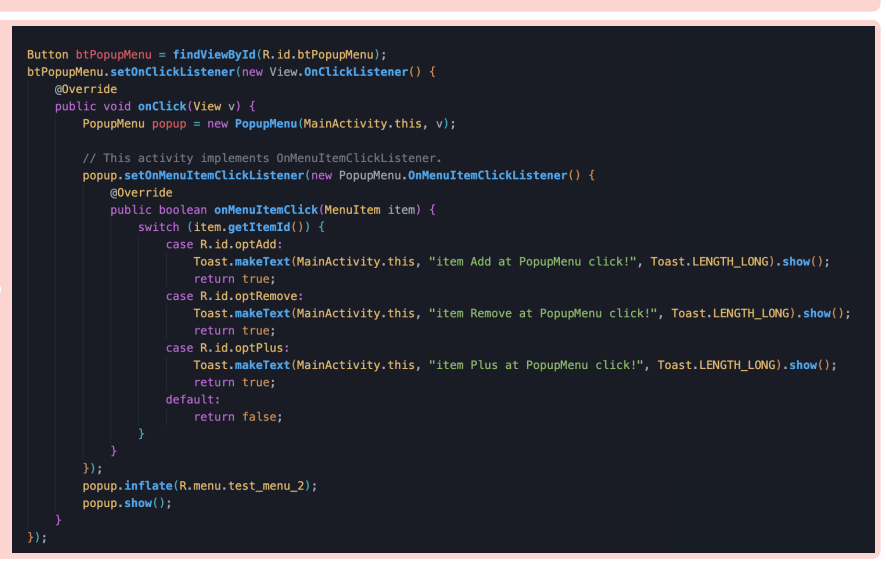
```java
Button btnContextMenuAction = findViewById(R.id.btContextualMenuActionMode);
btnContextMenuAction.setOnLongClickListener(new View.OnLongClickListener() {
    @Override
    public boolean onLongClick(View v) {
        v.startActionMode(actionModeCallback);
        return true;
    }
});
```

③ **Handle click event** display



Popup Menu

① **Define Menu XML**

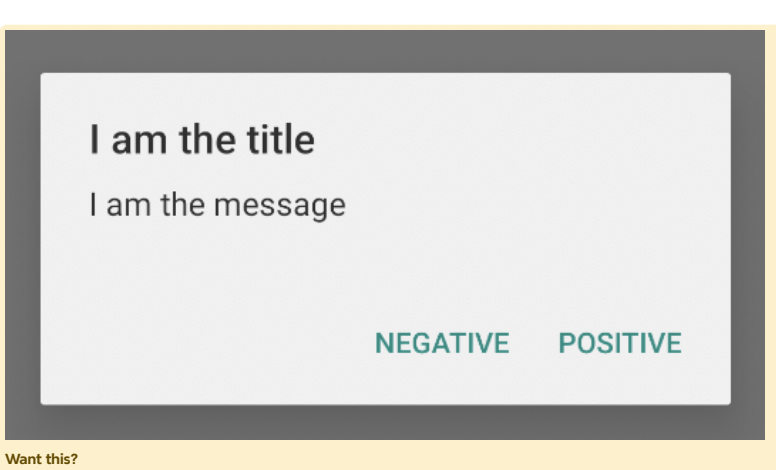


```java
Button btPopupMenu = findViewById(R.id.btPopupMenu);
btPopupMenu.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        PopupMenu popup = new PopupMenu(MainActivity.this, v);

        // This activity implements OnMenuItemClickListener.
        popup.setOnMenuItemClickListener(new PopupMenu.OnMenuItemClickListener() {
            @Override
            public boolean onMenuItemClick(MenuItem item) {
                switch (item.getItemId()) {
                    case R.id.optAdd:
                        Toast.makeText(MainActivity.this, "item Add at PopupMenu click!", Toast.LENGTH_LONG).show();
                        return true;
                    case R.id.optRemove:
                        Toast.makeText(MainActivity.this, "item Remove at PopupMenu click!", Toast.LENGTH_LONG).show();
                        return true;
                    case R.id.optPlus:
                        Toast.makeText(MainActivity.this, "item Plus at PopupMenu click!", Toast.LENGTH_LONG).show();
                        return true;
                    default:
                        return false;
                }
            }
        });
        popup.inflate(R.menu.test_menu_2);
        popup.show();
    }
});
```

②

Dialog

Alert Dialog

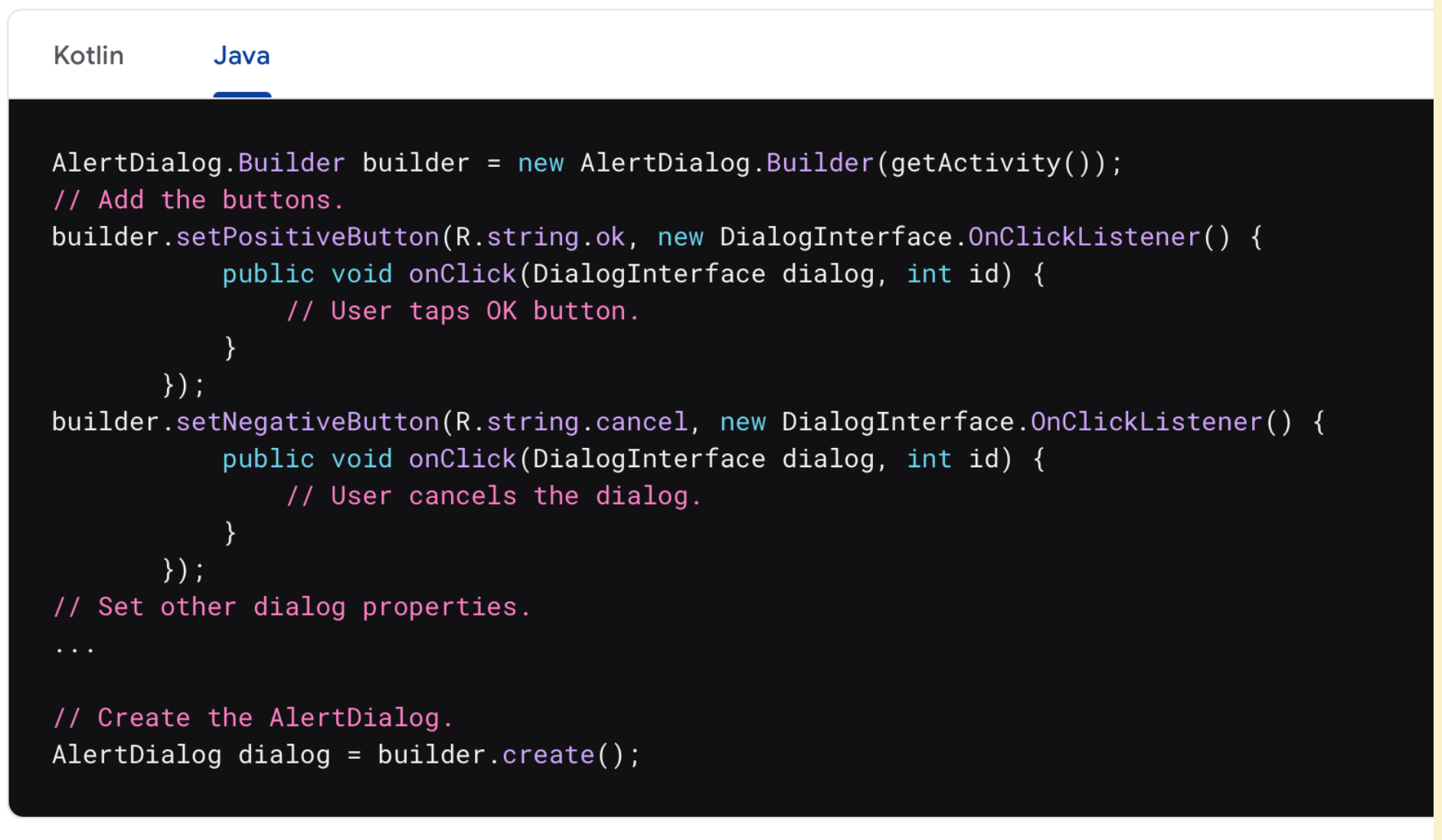


I am the title

I am the message

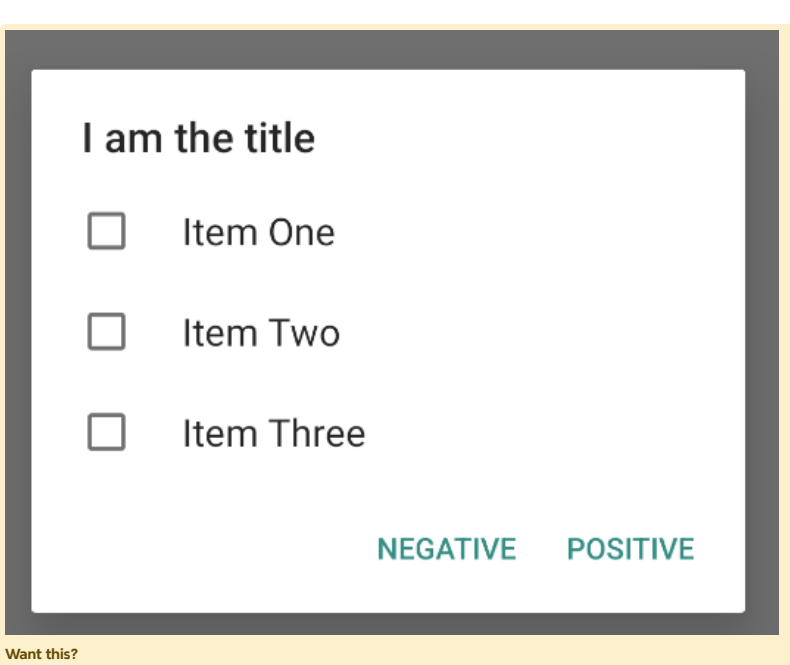
NEGATIVE POSITIVE

Want this?



```java
AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());
// Add the buttons.
builder.setPositiveButton(R.string.ok, new DialogInterface.OnClickListener() {
           public void onClick(DialogInterface dialog, int id) {
               // User taps OK button.
           }
       });
builder.setNegativeButton(R.string.cancel, new DialogInterface.OnClickListener() {
           public void onClick(DialogInterface dialog, int id) {
               // User cancels the dialog.
           }
       });
// Set other dialog properties.
...

// Create the AlertDialog.
AlertDialog dialog = builder.create();
```

Do this!



I am the title

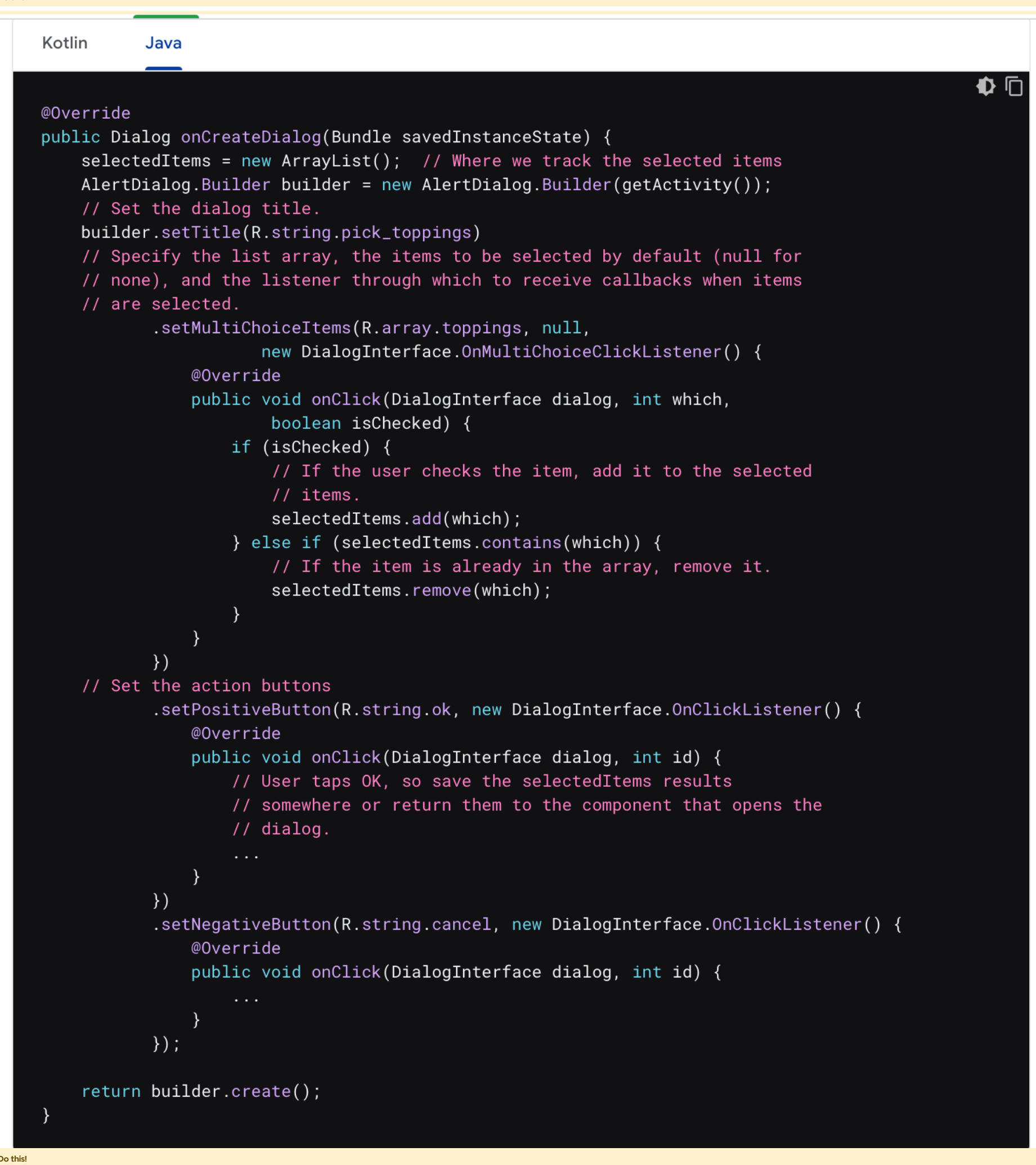
☐ Item One

☐ Item Two

☐ Item Three

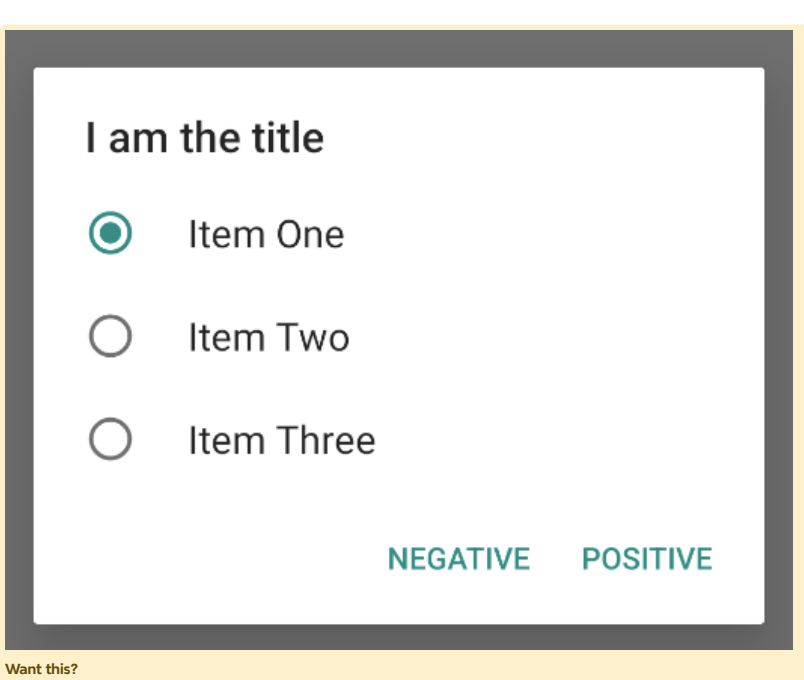
NEGATIVE POSITIVE

Want this?



```java
@Override
public Dialog onCreateDialog(Bundle savedInstanceState) {
    selectedItems = new ArrayList();  // Where we track the selected items
    AlertDialog.Builder builder = new AlertDialog.Builder(getActivity());
    // Set the dialog title.
    builder.setTitle(R.string.pick_toppings)
    // Specify the list array, the items to be selected by default (null for
    // none), and the listener through which to receive callbacks when items
    // are selected.
           .setMultiChoiceItems(R.array.toppings, null,
                      new DialogInterface.OnMultiChoiceClickListener() {
               @Override
               public void onClick(DialogInterface dialog, int which,
                       boolean isChecked) {
                   if (isChecked) {
                       // If the user checks the item, add it to the selected
                       // items.
                       selectedItems.add(which);
                   } else if (selectedItems.contains(which)) {
                       // If the item is already in the array, remove it.
                       selectedItems.remove(which);
                   }
               }
           })
    // Set the action buttons
           .setPositiveButton(R.string.ok, new DialogInterface.OnClickListener() {
               @Override
               public void onClick(DialogInterface dialog, int id) {
                   // User taps OK, so save the selectedItems results
                   // somewhere or return them to the component that opens the
                   // dialog.
                   ...
               }
           })
           .setNegativeButton(R.string.cancel, new DialogInterface.OnClickListener() {
               @Override
               public void onClick(DialogInterface dialog, int id) {
                   ...
               }
           });

    return builder.create();
}
```

Do this!



I am the title

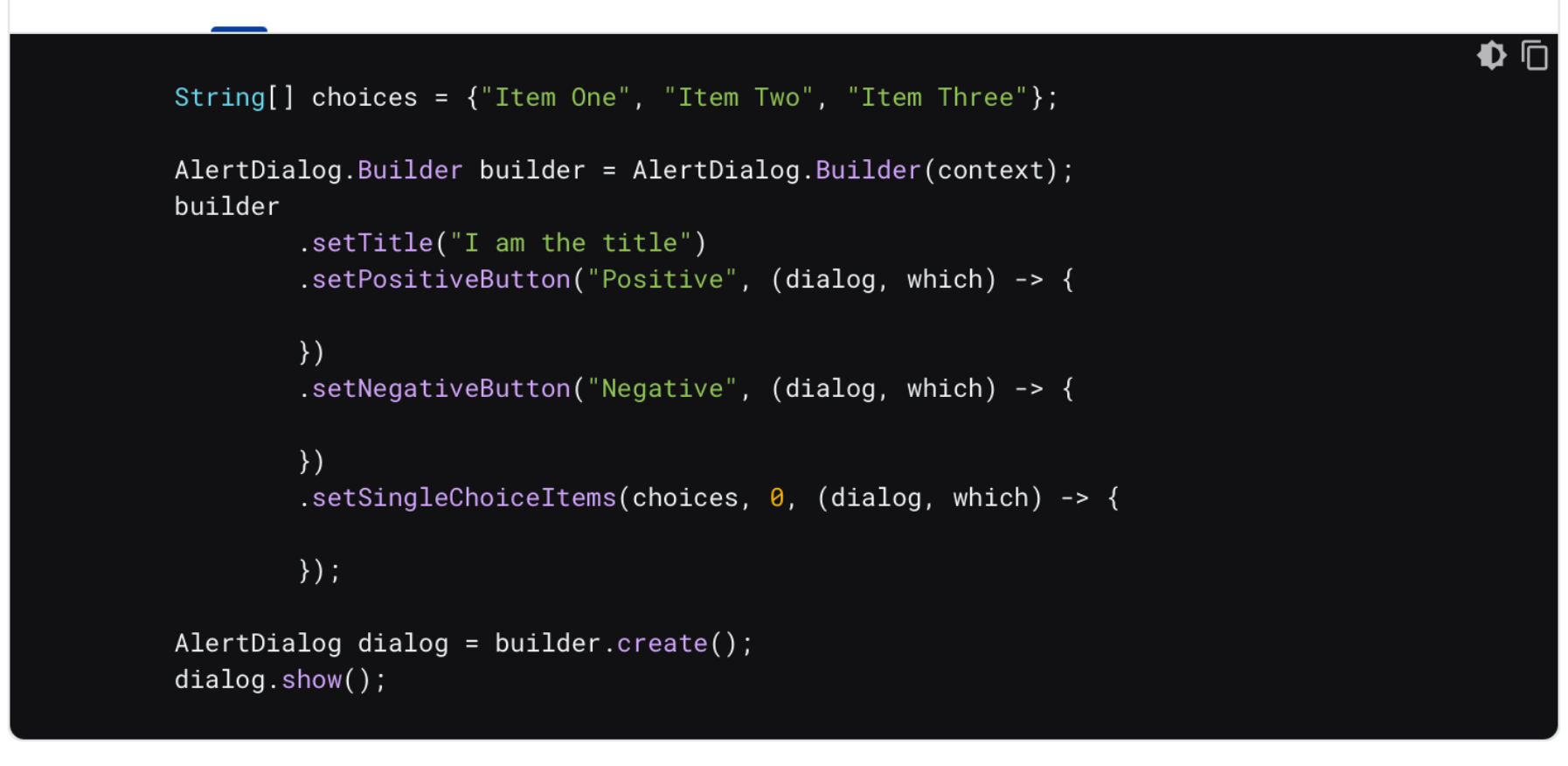
● Item One

○ Item Two

○ Item Three

NEGATIVE POSITIVE

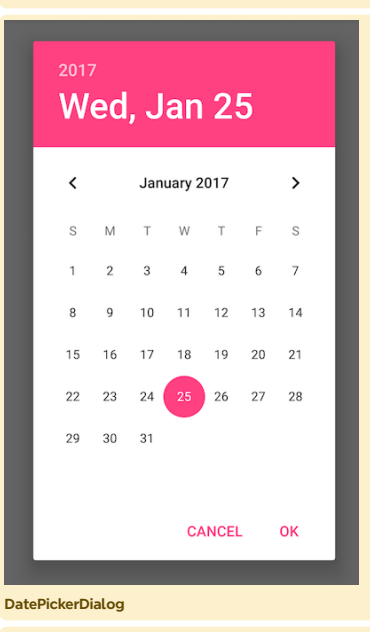
Want this?



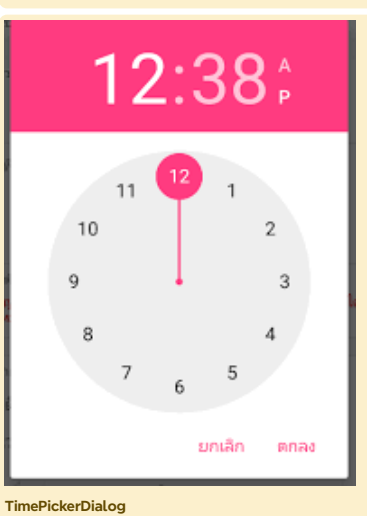
```java
        String[] choices = {"Item One", "Item Two", "Item Three"};

        AlertDialog.Builder builder = AlertDialog.Builder(context);
        builder
                .setTitle("I am the title")
                .setPositiveButton("Positive", (dialog, which) -> {

                })
                .setNegativeButton("Negative", (dialog, which) -> {

                })
                .setSingleChoiceItems(choices, 0, (dialog, which) -> {

                });

        AlertDialog dialog = builder.create();
        dialog.show();
```

Do this!

How to custom your own dialog style --------------->



DatePickerDialog



TimePickerDialog